

IT PROJECT MANAGEMENT

TOPIC 13

MONITORING AND CONTROLLING PROJECTS

LEARNING OBJECTIVES

a. List several processes and outputs of project monitoring and controlling, and describe outputs common to all knowledge areas when using a predictive approach to project management
b. Discuss monitoring and controlling project work and performing integrated change control as part of project integration management and how to use earned value management
c. Explain the importance of validating and controlling scope
d. Describe the schedule control process and schedule performance measurement tools, such as tracking Gantt charts
e. Discuss tools and techniques to assist in cost control
f. List the Seven Basic Tools of Quality, and provide examples of how they assist in performing quality control
g. Explain the process of controlling resources
h. Summarize methods for monitoring communications
i. Discuss different approaches to monitoring stakeholder engagement
j. Describe the process of monitoring risks
k. Explain how to control procurements
l. Apply the project management principles to monitoring and controlling projects
m. Discuss unique aspects of monitoring and controlling agile/hybrid projects

MONITORING AND CONTROLLING PROCESSES

GOALS

- Monitoring and controlling involves regularly **measuring progress** to ensure that the project is meeting its objectives and addressing current business needs.
- The project manager and other staff monitor progress against plans and take **corrective** action when necessary.

SUMMARY OF MONITORING AND CONTROLLING PROCESSES AND OUTPUTS

Knowledge area	Monitoring and controlling process	Outputs
Project integration management	Monitor and control project work Perform integrated change control	Work performance reports Change requests Project management plan updates Project documents updates Approved change requests Project management plan updates Project documents updates
Project scope management	Validate scope Control scope	Accepted deliverables Work performance information Change requests Project documents updates Work performance information Change requests Project management plan updates Project documents updates
Project schedule management	Control schedule	Work performance information Schedule forecasts Change requests Project management plan updates Project documents updates
Project cost management	Control cost	Work performance information Cost forecasts Change requests Project management plan updates Project documents updates
Project quality management	Control quality	Quality control measurements Verified deliverables Work performance information Change requests Project management plan updates Project documents updates
Project resource management	Control resources	Work performance information Change requests Project management plan updates Project documents updates
Project communications management	Monitor communications	Work performance information Change requests Project management plan updates Project documents updates
Project stakeholder management	Monitor stakeholder engagement	Work performance information Change requests Project management plan updates Project documents updates
Project risk management	Monitor risks	Work performance information Change requests Project management plan updates Project documents updates Organizational process assets updates
Project procurement management	Control procurements	Closed procurements Work performance information Procurement documentation updates Change requests Project management plan updates Project documents updates Organizational process assets updates

PROJECT INTEGRATION MANAGEMENT

MONITORING AND CONTROLLING PROJECT WORK

- **Project changes are inevitable**, so it is important to develop and follow a process to monitor and control them.
- Monitoring and controlling project work includes collecting, measuring, and disseminating performance information.
  - The project team should continuously monitor project performance to assess the overall health of the project and identify areas that require special attention.
- The project management plan, project documents, work performance reports (including earned value management and performance reports), agreements, enterprise environmental factors, and organizational process assets are all important inputs for monitoring and controlling project work.
- The main tools and techniques for monitoring and controlling project work are *expert judgment, data analysis, decision making, and meetings.*
- **Earned value management** reports are a powerful tool for monitoring and controlling overall project performance.

FORECASTING WITH EARNED VALUE MANAGEMENT

- **Earned value management (EVM)** is a project performance measurement technique that integrates scope, time, and cost data.
- Given a baseline, project managers and their teams can **determine how well the project** is meeting scope, time, and cost goals by entering actual information and then comparing it to the **baseline**.
  - A **baseline** is a starting point, a measurement, or an observation that is documented so that it can be used for future comparison.
- In earned value management, a baseline includes the following:
  - Scope (WBS deliverables)
  - Time (start and finish estimates for each activity)
  - Cost information (cost estimates for each activity)

EVM : PV, AC, EV

Earned value management involves determining three values for each deliverable or summary deliverable from a project's WBS :

1. The **planned value (PV)** is the authorized budget assigned to scheduled work.

   The cost baseline for the Just-In-Time Training project included $5,000 to be spent on course development for supplier management training. If the activity involved delivering a detailed course outline to be finished in one week for $5,000, then the planned value (PV) for that activity that week would be $5,000.

2. The **actual cost (AC)** is the realized cost incurred for the work performed on an activity during a specific time period

   For example, suppose it actually took one week and cost $6,000 to create the detailed course outline because the hourly rate for the person doing the work was higher than planned. The actual cost (AC) for the activity would therefore be $6,000.

3. The **earned value (EV)** is the measure of work performed expressed in terms of the budget authorized for that work.

   It cannot be greater than the authorized PV budget for a component as it is calculated as the sum of the PV of the completed work. For this example, the EV is $5,000.

EARNED VALUE FORMULA

- In general, **negative numbers** for cost and schedule variance **indicate problems** in those areas.
- Negative numbers mean the project is costing more than planned (over budget) or taking longer than planned (behind schedule).
- Likewise, CPI and SPI less than one or less than 100% indicate problems.

Term	Formula
Earned value (EV)	EV = PV of all completed work
Cost variance (CV)	CV = EV − AC
Schedule variance (SV)	SV = EV − PV
Cost performance index (CPI)	CPI = EV/AC
Schedule performance index (SPI)	SPI = EV/PV
Estimate at completion (EAC)	EAC = BAC/CPI
Estimated to Complete (ETC)	ETC = EAC − AC

EVM GRAPH

- The **budget at completion (BAC)**, or the approved total budget for the project, can be divided by the cost performance index to calculate the **estimate at completion (EAC)**, which is a forecast of how much the project will cost upon completion.
  - This calculation assumes you will be spending at the same rate as your current level of spending.
- Likewise**, the approved time estimate** for the project can be divided by the schedule performance index to calculate when the project will be completed.
- Earned value, therefore, provides an excellent way to monitor project performance **and provide forecasts based on performance to date**.

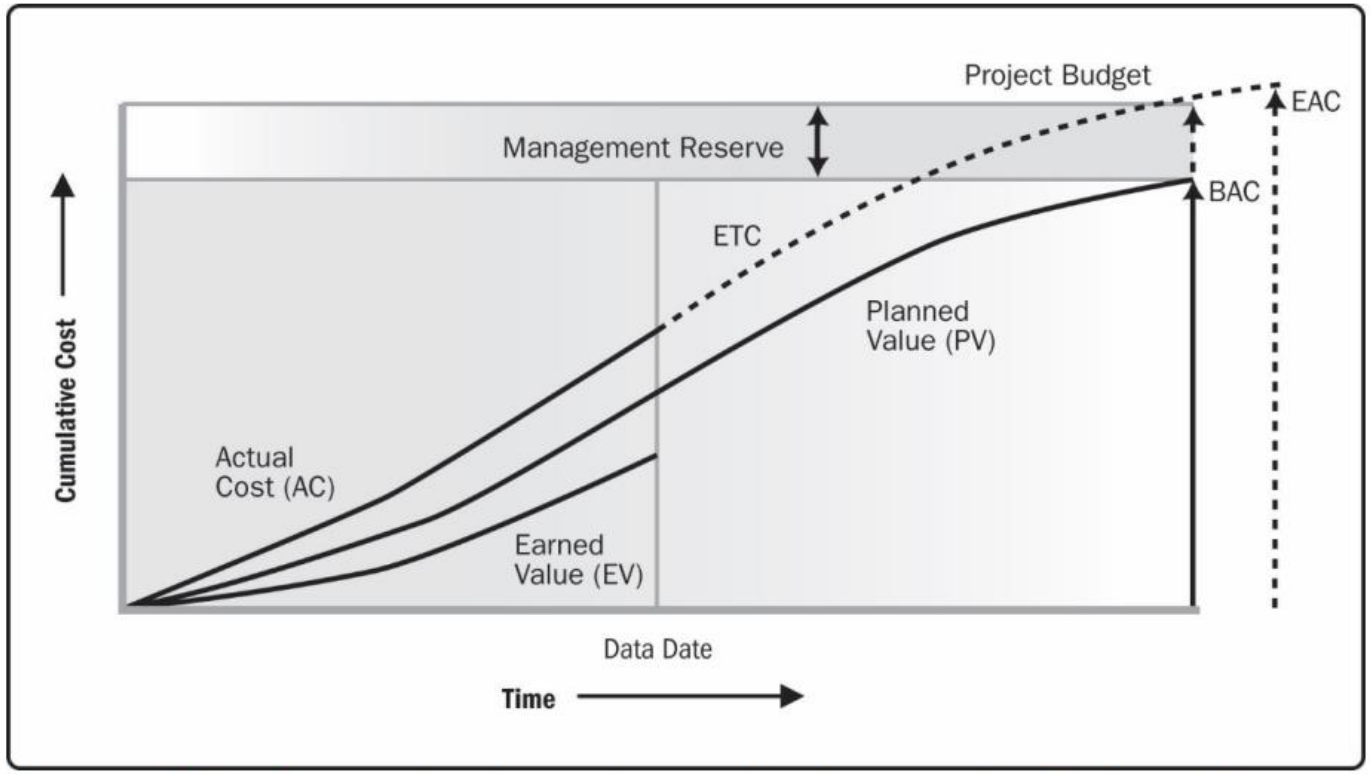
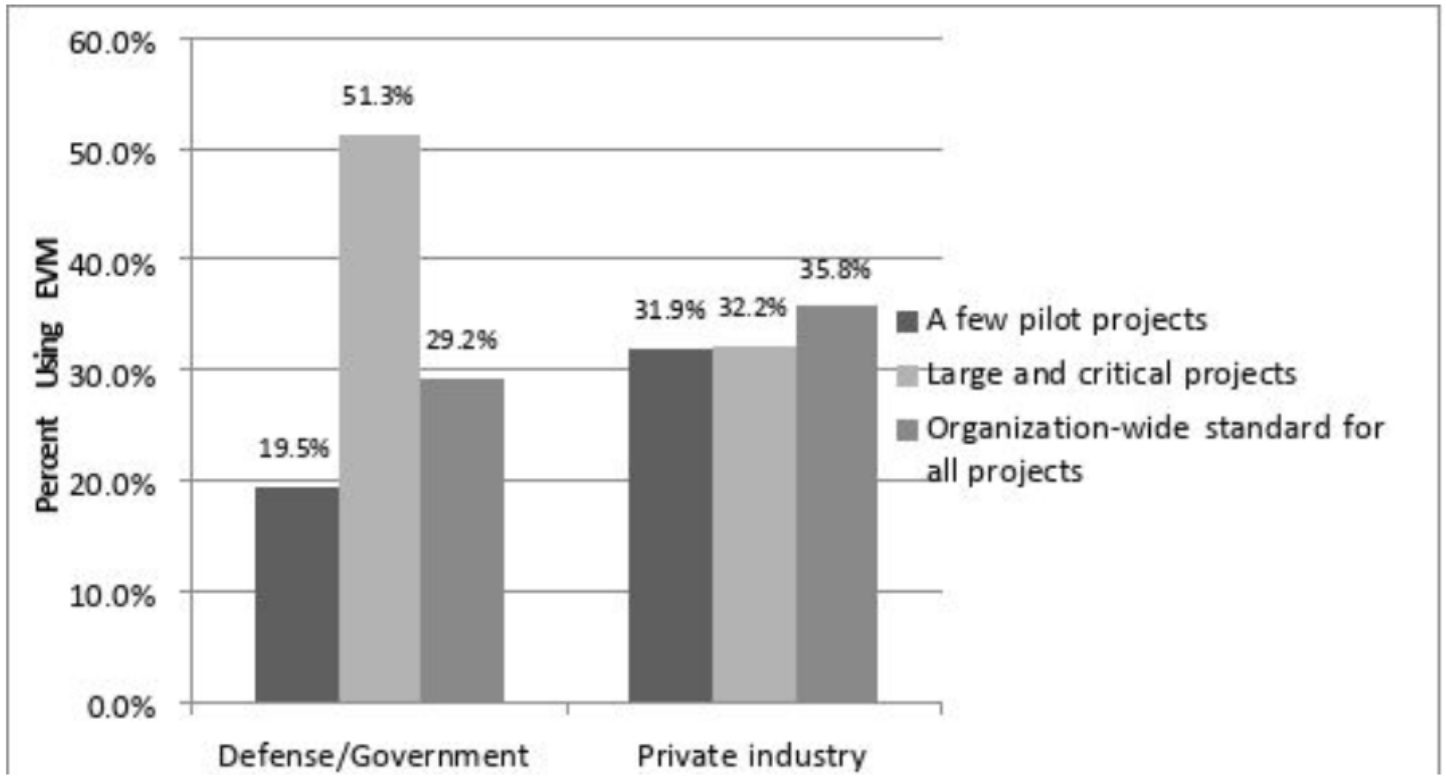


Figure 8-2.b. Earned value, planned value, and actual cost
Source: Project Management Institute, Inc., *A Guide to the Project Management Body of Knowledge (PMBOK® Guide) – Sixth Edition* (2017), Figure 7-12, Page 264.

EVM PRACTICE (2011)

- EVM is used worldwide, and it is popular in the Middle East, South Asia, Canada, and Europe.
- Most countries require EVM for large defense or government projects.
- Project budget size appears to be the most important factor in deciding whether or not to use EVM.



Source: Lingguang Song, "Earned Value Management: A Global and Cross-Industry Perspective on Current EVM Practice," PMI (2011).

Figure 8-4. Percentage of organizations using EVM

EXAMPLE - FORECAST USING AN EARNED VALUE CHART

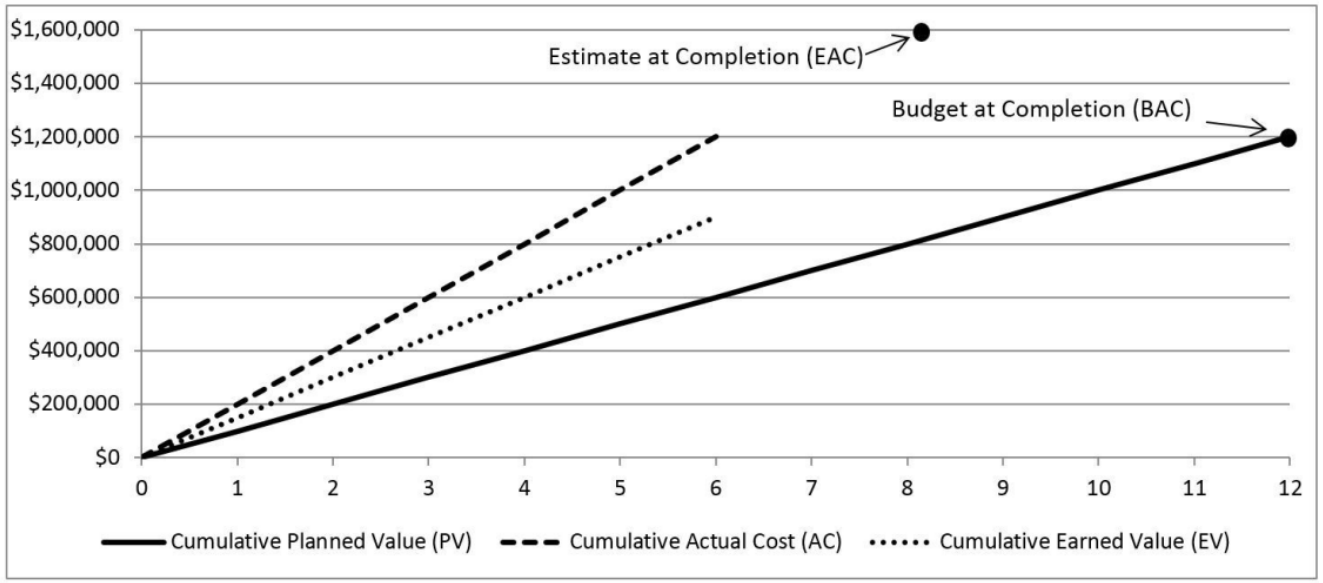
- The budget at completion is $1.2 million for this one-year project. The BAC point on the chart, therefore, is at 12 months and $1.2 million. Based on data for months one through six, an earned value chart was created. In this example, the planned value is $100,000 for each month, so the cumulative planned value at month 6 is $600,000. The earned value is $150,000 each month, with the cumulative earned value at month 6 of $900,000. The actual cost is $200,000 each month, with the cumulative actual cost at month 6 of $1,200,000

- CPI = EV/AC = $900,000/$1,200,000 = .75
- SPI = EV/PV = $900,000/$600,000 = 1.5
- EAC = BAC/CPI = $1,200,000/.75 = $1,600,000
- New completion date estimate = Original time estimate/ SPI = 12 months/1.5 = 8 months*

- *Note that some experts think SPI should not be used for estimating new completion dates because you need to reevaluate the critical path of the project

EXAMPLE - FORECAST USING AN EARNED VALUE CHART

- If the project goes exactly as planned, it will finish in 12 months and cost $1,200,000, as represented by the BAC point.
- Notice in the example that the actual cost line is always above the earned value line.
  - When the actual cost line is above the earned value line, costs are more than planned. That situation is usually not good.
- The planned value line in this example is always below the earned value line.
  - This relationship means that the project has been ahead of schedule the first six months. That situation is normally good, but it may not be.
- The forecasted completion date, therefore, is earlier than planned while the forecasted total cost is higher than planned.
  - As this trend started at month one and continued each month, you can assume that the project sponsor did want the project completed early and was willing to pay the extra cost. If this was not the case and the sponsor was not willing to pay more for an early completion, the project manager would have to scramble to try to complete the project within the approved time and budget constraints.
- The estimate at completion (EAC) is an important input to budget decisions, especially if total funds are limited.



PERFORMANCE REPORTS

- **Status reports** describe where the project stands at a specific point in time.
  - Recall the importance of the triple constraint. Status reports address where the project stands in terms of meeting scope, time, and cost goals. Is work being accomplished as planned? How long did it take to do certain activities? How much money has been spent to date? Status reports can take various formats depending on the stakeholders' needs.
- **Progress reports** describe what the project team has accomplished during a certain period.
  - In many projects, each team member prepares a weekly or monthly progress report. Team leaders often create consolidated progress reports based on the information received from team members.
- **Forecasts** predict future project status and progress based on past information and trends.
  - How long will it take to finish the project based on how things are going? How much more money will be needed to complete the project? Project managers can also use earned value management, as described in the previous section, to answer these questions by estimating the budget at completion and the projected completion date based on how the project is progressing.

SAMPLE PERFORMANCE REPORT

Progress Report

Project Name: Just-In-Time Training Project
Project Manager Name: Kristin Maur
Date: February 3
Reporting Period: January 1 – February 1

Work completed this reporting period:

- Held first negotiating skills course (instructor-led) with 20 participants
- Held first supplier management executive course (instructor-led) with 17 participants
- Held second supplier management introductory course (instructor-led) with 20 participants
- Had 32 people begin the Web-based introductory supplier management course
- Continued developing other Web-based courses
- Prepared evaluations of all courses held to date

Work to complete next reporting period:

- Hold first advanced supplier management course
- Hold first project management course
- Hold first software applications course

What's going well and why:

- Participation in all courses is good. Every instructor-led course was full, except the supplier management executive course. All the courses were advertised well, and we had more than enough people sign-up for the classes. We put several people on the list for later courses after courses were filled in the registration system.
- The average course ratings were above 3.8 on a 5.0 scale. Comments were generally very positive.
- More people than expected started the first Web-based course. Development of new Web-based courses is going well.

What's not going well and why:

- We did not fill the supplier management executive course as planned. Three people could not attend at the last minute, and it was too late to get replacements. We will work on a policy to help prevent this problem in the future for all instructor-led classes.
- We were surprised that so many people started the Web-based introductory supplier management course. We can handle the numbers, but we could have done a better job at forecasting demand.

Suggestions/Issues:

- Develop a policy to handle people not being able to attend classes at the last minute.
- Try to do a better job at forecasting demand for Web-based courses.

Project changes:

No major changes to report. The earned value chart in Attachment 1 shows planned value, actual cost, and earned value information to date. We are very close to our plans, running slightly ahead of schedule and a bit over budget.

Figure 8-6. Sample performance report

INTEGRATED CHANGE CONTROL

Integrated change control involves identifying, evaluating, and managing changes throughout the project's life cycle. The three main objectives of integrated change control are as follows:

- 1. *Influencing the factors that cause changes to ensure that changes are beneficial*: Changes can often be good for a project, so it is important to let everyone know that and focus on promoting changes that are beneficial.
  - For example, changes that improve quality, reduce costs, save time, or improve stakeholder relationships are beneficial.
- 2. *Determining that a change has occurred*: To determine that a change has occurred, project managers must know the status of key project areas at all times.
- 3. *Managing actual changes as they occur*: Managing change is a key role of project managers and their teams.

PROJECT SCOPE MANAGEMENT

- The main monitoring and controlling processes performed as part of project scope management are **validating scope and controlling scope.**
- Key outputs are **accepted deliverables and work performance information**.
- Detailed procedures must be developed to ensure that customers are getting what they want and that the project team has enough time and money to produce the desired products and services.
- Even when the project scope is fairly well defined, many projects suffer from **scope creep**

VALIDATING SCOPE

- **Scope validation** involves formal acceptance of the completed project deliverables by the project customer or designated stakeholders.
- This acceptance is often achieved **through customer inspection and then sign-off on key deliverables.**
- To receive formal acceptance of the project scope, the project team must develop clear documentation of the project's products and procedures, which the appropriate stakeholders can then evaluate for completion and their satisfaction with the results.
- The project management plan, project documents, verified deliverables, and work performance data are the **main inputs** for scope validation.
- A verified deliverable has been completed and checked for correctness as part of quality control, while **validation means acceptance by the customer or identified stakeholders.**
- The **main tools** for performing scope validation are inspection and decision-making.
  - The customer inspects the work after it is delivered to decide if it is acceptable. The customer is often more than one person, so group decision-making is often required for the inspection and acceptance.

SAMPLE OF ACCEPTED AND UNACCEPTED DELIVERABLES

Deliverable Acceptance Form

Project Name: Just-In-Time Training Project **Date:** November 12

Deliverable Name: Course materials for introductory supplier management course

Project Manager: Kristin Maur **Project Sponsor:** Lucy Camerena

(We), the undersigned, acknowledge and accept delivery of the work completed for this deliverable on behalf of our organization. My (Our) signature(s) attest(s) to agreement that this deliverable has been completed. No further work should be done on this deliverable. If the deliverable is not acceptable, reasons are stated and corrective actions are described.

1. Was this deliverable completed to your satisfaction? Yes _____No X
2. Please provide the main reasons for your satisfaction or dissatisfaction with this deliverable.

As stated in the contract statement of work, the course materials are not completed until all constructive feedback from the prototype course has been incorporated or the supplier has provided strong rationale as to why the feedback should not be incorporated. We requested that a new section be added to the course to cover issues related to working with suppliers in virtual settings. The final materials delivered did not include this new section or discuss why it was not added. We believe it was an oversight that can be corrected with a minimal amount of additional work.

3. If the deliverable is not acceptable, describe in detail what additional work must be done.

The supplier will add a new section to the course on working with suppliers in a virtual setting. This section should take about thirty minutes of class time in a face-to-face or e-learning setting. It will follow the format and review process used for other topics in the course. We request the draft of this new section within one week and final delivery within two weeks.

Contact's signature for resubmission of deliverable if found unacceptable: *Kristin Maur*

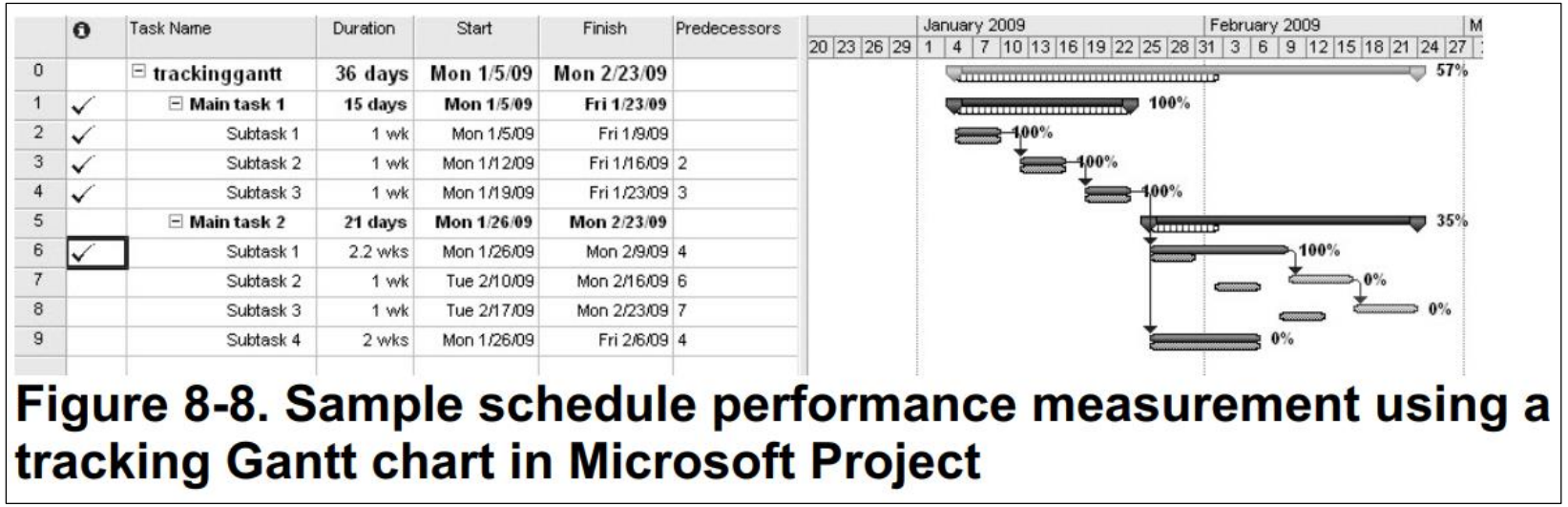
CONTROLLING SCOPE

- You cannot control the scope of a project unless you have first clearly defined the scope and set a scope validation process in place.
- You also need to develop a process for soliciting and monitoring changes to project scope.
- Stakeholders should be encouraged to suggest beneficial changes and discouraged from suggesting unnecessary changes

PROJECT SCHEDULE MANAGEMENT

- The main monitoring and controlling process performed as part of project schedule management is controlling the schedule or schedule control.
- Project managers often cite delivering projects on time (schedule control) as one of their biggest challenges, because schedule problems often cause more conflict than other issues.
- During project initiation, priorities and procedures are often most important, but as the project proceeds, especially during the middle and latter stages of a project, schedule issues become the predominant source of conflict.
- Perhaps part of the reason schedule problems are so common is that time is easily and simply measured.
- After a project schedule is set, anyone can quickly estimate schedule performance by subtracting the original time estimate from the time actually expended.
- People often compare planned and actual project completion times without considering the approved project changes.
- Time is also the variable with the least amount of flexibility.
- Time passes no matter what happens on a project.
- Individual work styles and cultural differences may also cause schedule conflicts. For example, one dimension of the MyersBriggs team-building tool that was described in Chapter 6 (Judgment/Perception, or J/P) deals with peoples' attitudes toward structure and deadlines. Some people (J's) prefer detailed schedules and focus on task completion. Others (P's) prefer to keep things open and flexible. Different cultures and even entire countries have different attitudes about schedules. For example, in some countries, businesses close for several hours every afternoon so workers can take naps. Others observe religious or secular holidays during which little work is accomplished. Cultures may also have different perceptions of work ethic—some may value hard work and strict schedules, whereas others may value the ability to remain relaxed and flexible.
- The goal of schedule control is to know the status of the schedule, influence the factors that cause schedule changes, determine whether the schedule has changed, and manage changes when they occur. Key outputs of schedule control include schedule forecasts and work performance information. There are several tools and techniques for schedule control, including data analysis (such as earned value management, as described earlier in this chapter), project management information systems, resource optimization, and several tools discussed in Chapter 5 (such as critical path method, leads and lags, and schedule compression).

SAMPLE WORK PERFORMANCE INFORMATION



	Task Name	Duration	Start	Finish	Predecessors
0	trackinggantt	36 days	Mon 1/5/09	Mon 2/23/09	
1	Main task 1	15 days	Mon 1/5/09	Fri 1/23/09	
2	Subtask 1	1 wk	Mon 1/5/09	Fri 1/9/09	
3	Subtask 2	1 wk	Mon 1/12/09	Fri 1/16/09	2
4	Subtask 3	1 wk	Mon 1/19/09	Fri 1/23/09	3
5	Main task 2	21 days	Mon 1/26/09	Mon 2/23/09	
6	Subtask 1	2.2 wks	Mon 1/26/09	Mon 2/9/09	4
7	Subtask 2	1 wk	Tue 2/10/09	Mon 2/16/09	6
8	Subtask 3	1 wk	Tue 2/17/09	Mon 2/23/09	7
9	Subtask 4	2 wks	Mon 1/26/09	Fri 2/6/09	4

Figure 8-8. Sample schedule performance measurement using a tracking Gantt chart in Microsoft Project

- Project managers often illustrate schedule progress at these meetings with a **tracking Gantt chart**—a Gantt chart that compares planned and actual project schedule information.
- Many project managers believe that tracking Gantt charts are an excellent tool for tracking project schedule performance and reporting that information to stakeholders.
- The tracking Gantt chart shows bars for both planned and actual start and finish dates for each task as well as the percent of work completed.

SAMPLE WORK PERFORMANCE INFORMATION

- Notice that the tracking Gantt chart in Figure 8-8 often shows two horizontal bars for tasks. The top horizontal bar represents the planned or baseline duration for each task.
- The bar below it represents the actual duration.
- A white diamond on the tracking Gantt chart represents a slipped milestone. A **slipped milestone** refers to a milestone activity that was completed later than originally planned.
- Percentages to the right of the horizontal bars display the percentage of work completed for each task.
- For example, 100% indicates that the task is finished, whereas 50% indicates that the task is still in progress and is 50% completed.
- In the columns to the left of the tracking Gantt chart, you can display baseline and actual start and finish dates

PROJECT COST MANAGEMENT

- Cost control includes monitoring cost performance, ensuring that only appropriate project changes are included in a revised cost baseline, and informing project stakeholders of authorized changes to the project that will affect costs.
- The project management plan, project documents, project funding requirements, work performance data, and organizational process assets are inputs to the cost-control process.
- Outputs of cost control include work performance information, cost forecasts, change requests, project management plan updates, and project documents updates. Several tools and techniques assist in project cost control, including expert judgment, data analysis, project management information systems, and the **To-complete performance index (TCPI)**, which is a measure of the cost performance that must be achieved on the remaining work in order to meet a specified goal, such as the BAC or EAC. As mentioned in the following What Went Right? examples, there is great room for improvement in project cost management in many industries.

PROJECT QUALITY MANAGEMENT

The main project quality management process for monitoring and controlling is quality control. Key outputs include quality-control measurements, verified deliverables, work performance information, change requests, project management plan updates, and project documents updates. Although one of the main goals of quality control is to ensure and improve quality, the main outcomes of this process are acceptance decisions, rework, and process adjustments.

1. *Acceptance decisions* determine if the products or services produced as part of the project will be accepted or rejected. If they are accepted, they are considered to be verified deliverables. If project stakeholders reject some of the products or services produced as part of the project, there must be rework.
2. *Rework* is action taken to bring rejected items into compliance with product requirements or specifications or other stakeholder expectations. Rework can be very expensive, so the project manager who excels at quality planning and quality assurance can reduce the need for rework.
3. *Process adjustments* correct or prevent further quality problems. Based on the implementation of quality-control measurements, process adjustments often result in updates to the quality baseline, organizational process assets, and the project management plan.

SAMPLE QUALITY-CONTROL TOOLS

Many different tools and techniques are available for performing quality control. The *PMBOK® Guide – Sixth Edition*, lists the following:

- Data gathering: Checklists, check sheets, statistical sampling, questionnaires, and surveys
- Data analysis: Performance reviews and root cause analysis Inspection
- Testing/product evaluations Data representation: cause-and-effect diagrams, control charts, histograms, and scatter diagrams
- Meetings

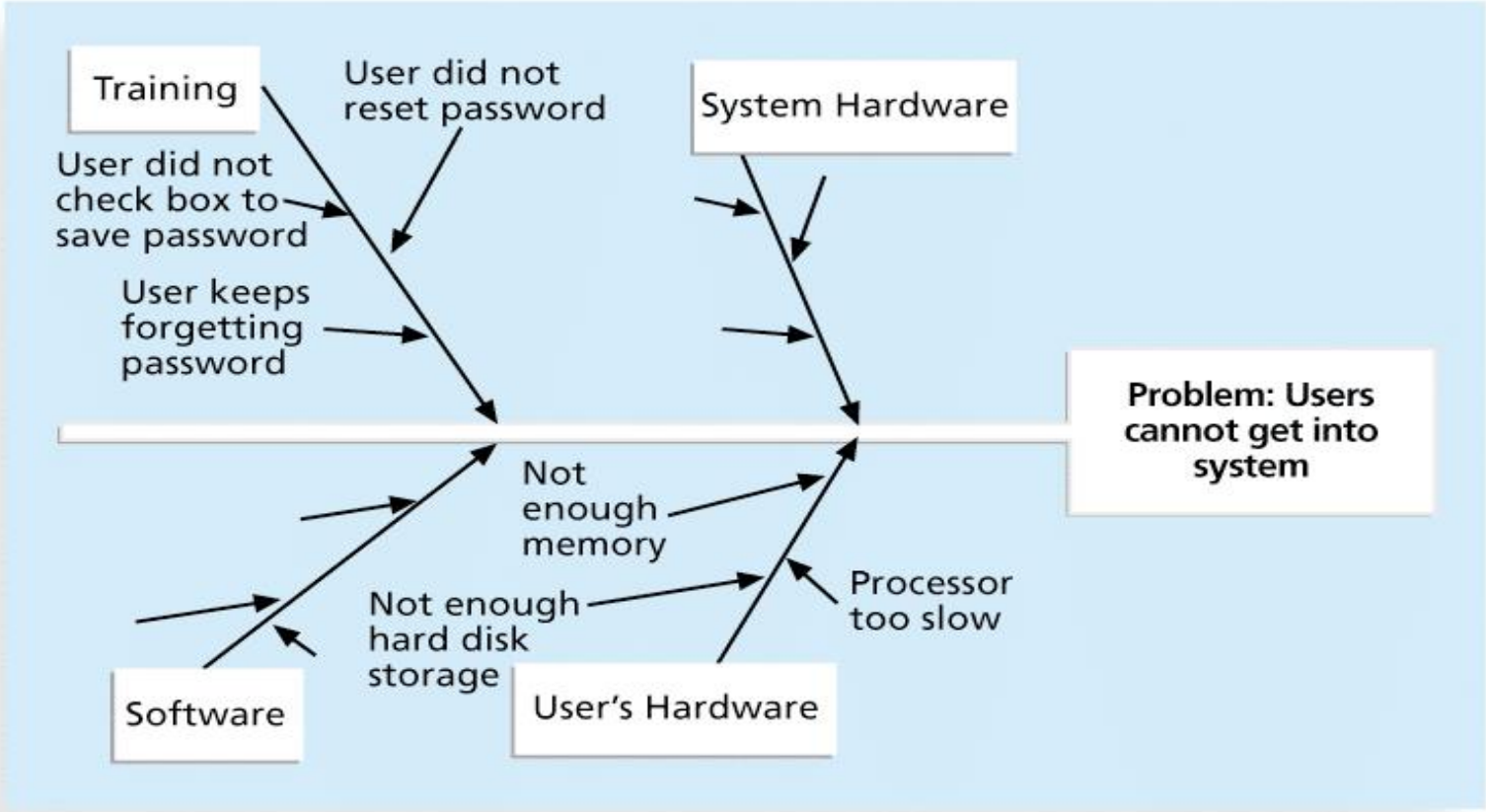
SEVEN BASIC TOOLS OF QUALITY

1. Cause-and-effect diagrams
2. Quality Control Charts
3. Checksheet
4. Scatter diagram
5. Histograms
6. Pareto Charts
7. Flowcharts

1. CAUSE-AND-EFFECT DIAGRAMS

- **Cause-and-effect diagrams** trace complaints about quality problems back to the responsible production operations
- They help you find the root cause of a problem
- Also known as **fishbone** or **Ishikawa diagrams**
- Can also use the **5 whys** technique where you repeated ask the question “Why” (five is a good rule of thumb) to peel away the layers of symptoms that can lead to the root cause

FIGURE 8-2. SAMPLE CAUSE-AND-EFFECT DIAGRAM

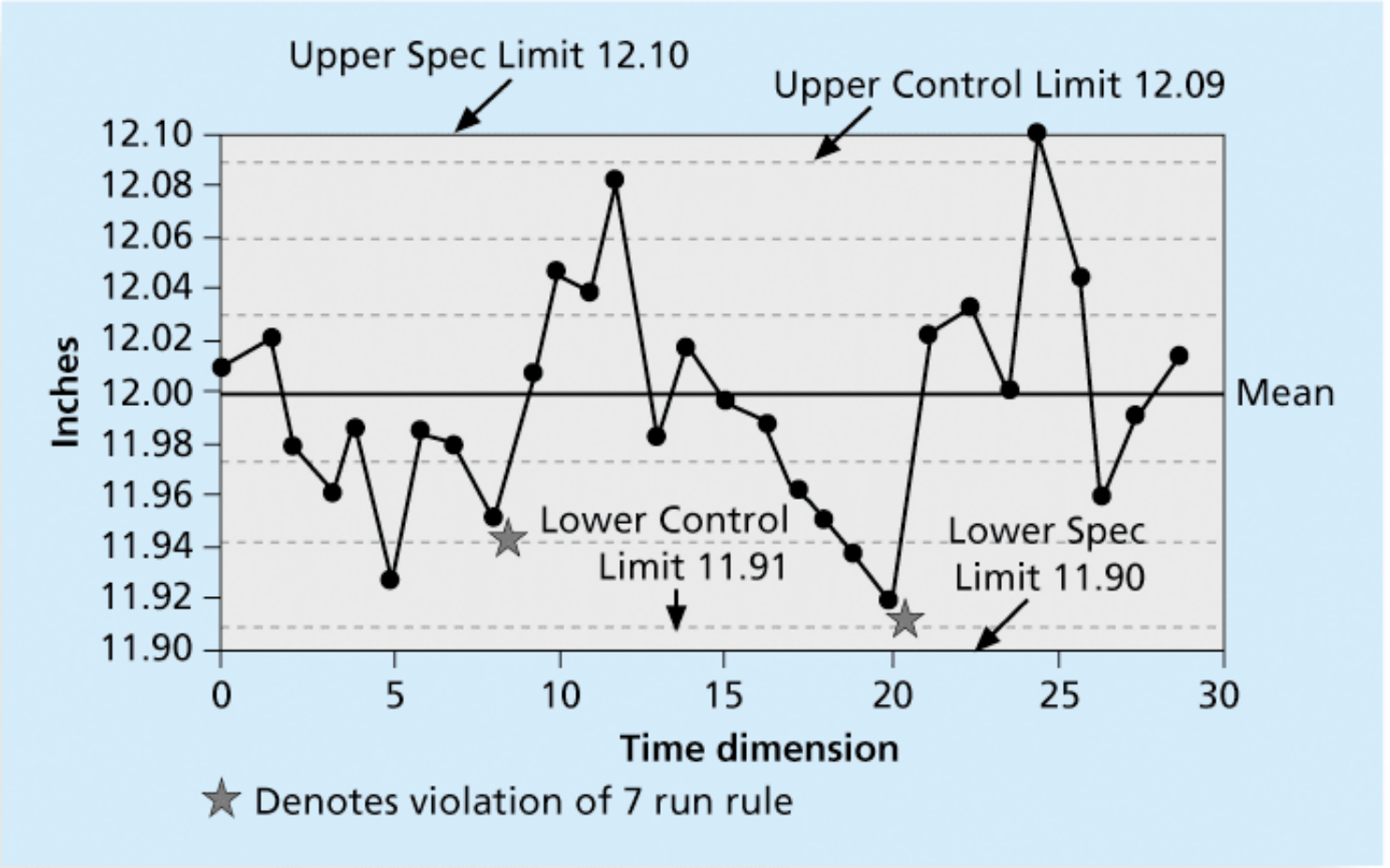


2. QUALITY CONTROL CHARTS

- A **control chart** is a graphic display of data that illustrates the results of a process over time
- The main use of control charts is to prevent defects, rather than to detect or reject them
- Quality control charts allow you to determine whether a process is in control or out of control
  - When a process is in control, any variations in the results of the process are created by random events; processes that are in control do not need to be adjusted
  - When a process is out of control, variations in the results of the process are caused by non-random events; you need to identify the causes of those non-random events and adjust the process to correct or eliminate them

THE SEVEN RUN RULE

- You can use quality control charts and the seven run rule to look for patterns in data
- The **seven run rule** states that if seven data points in a row are all below the mean, above the mean, or are all increasing or decreasing, then the proces needs to be examined for non-random problems



3. CHECKSHEET

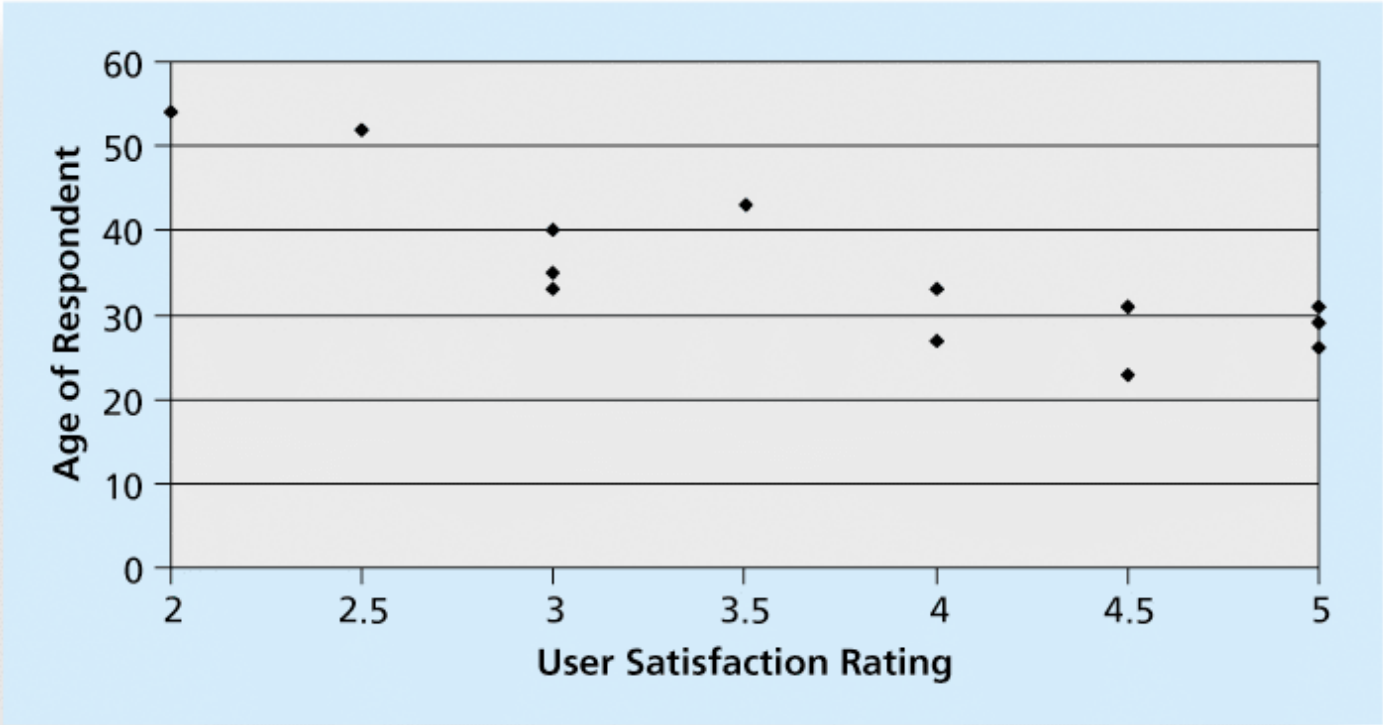
- A checksheet is used to collect and analyze data
- It is sometimes called a tally sheet or checklist, depending on its format
- In the example in Figure 8-4, most complaints arrive via text message, and there are more complaints on Monday and Tuesday than on other days of the week
- This information might be useful in improving the process for handling complaints

System Complaints

Source	Day							
	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday	Total
Email	𝍸	𝍸 |	|	||	|		|	12
Text	𝍸 ||	𝍸 ||	𝍸 |	|||	𝍸 |	||	|||	29
Phone call	|	||	|	|	||	|		8
Total	11	10	8	6	7	3	4	49

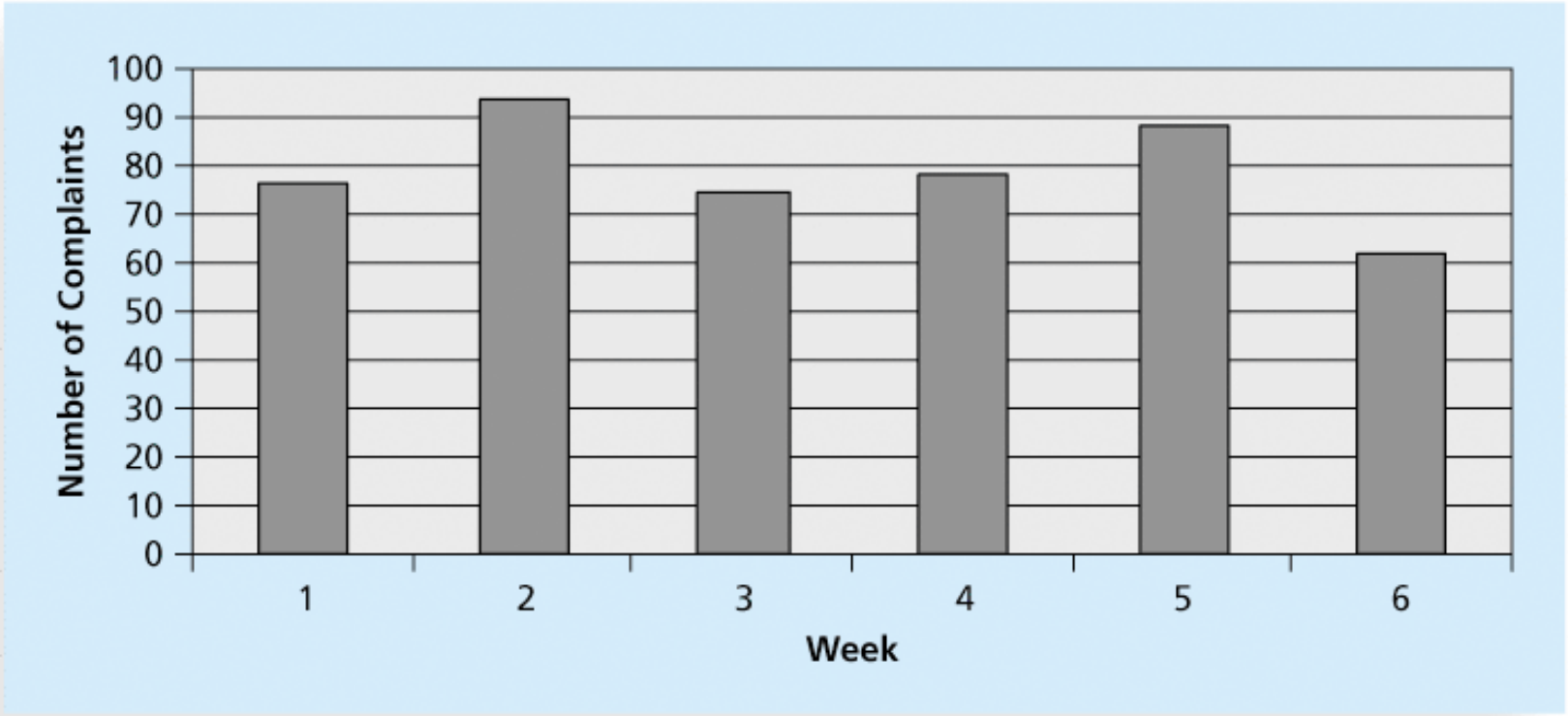
4. SCATTER DIAGRAM

- A **scatter diagram** helps to show if there is a relationship between two variables
- The closer data points are to a diagonal line, the more closely the two variables are related



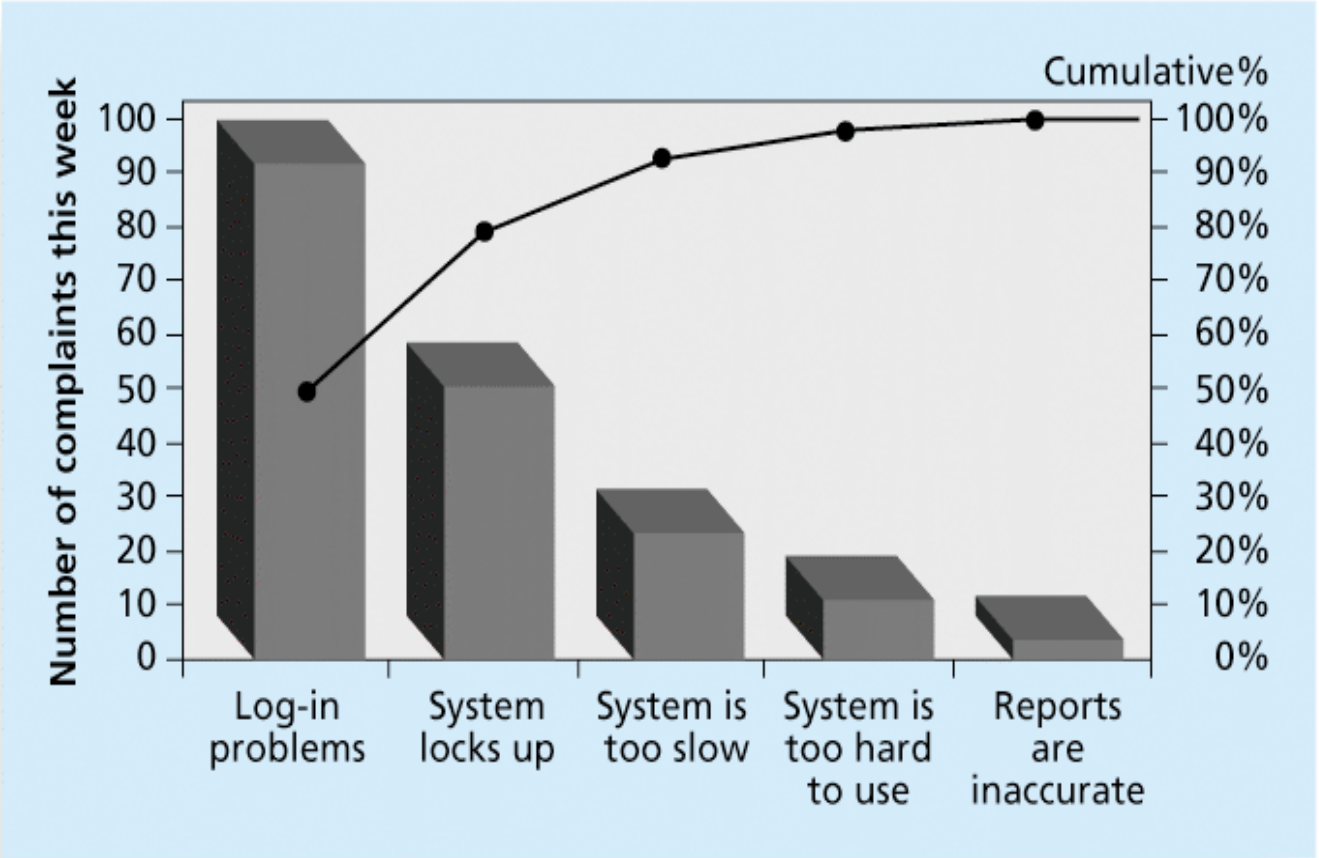
5. HISTOGRAMS

- A **histogram** is a bar graph of a distribution of variables
- Each bar represents an attribute or characteristic of a problem or situation, and the height of the bar represents its frequency



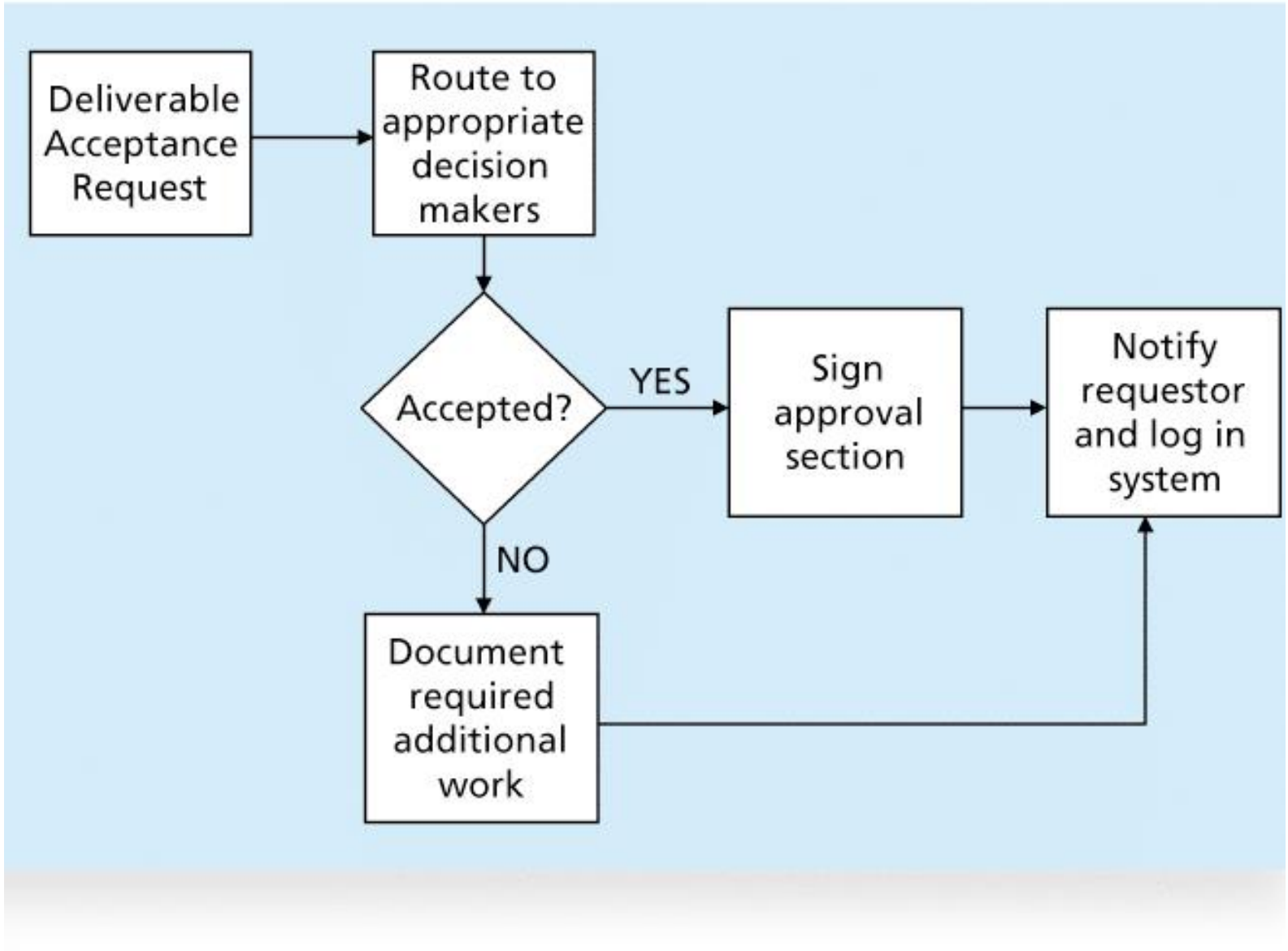
6. PARETO CHARTS

- A **Pareto chart** is a histogram that can help you identify and prioritize problem areas
- **Pareto analysis** is also called the 80-20 rule, meaning that 80 percent of problems are often due to 20 percent of the causes



7. FLOWCHARTS

- Flowcharts are graphic displays of the logic and flow of processes that help you analyze how problems occur and how processes can be improved
- They show activities, decision points, and the order of how information is processed



PROJECT RESOURCE MANAGEMENT

- Controlling resources involves **ensuring that the *physical* resources assigned to the project are available as planned** and monitoring the planned versus actual resources utilization, **taking corrective** actions as needed.
- Making effective use of team members is addressed under the Manage Team process.
- Tools and techniques include data analysis, problem solving, interpersonal and team skills, and project management information systems.
- Key outputs include work performance information, change requests, project management plan updates, and project document updates.

PROJECT COMMUNICATIONS MANAGEMENT

- Monitoring communications involves monitoring communications throughout the project life cycle to **ensure that stakeholder information needs are met**.
- Key *outputs* include work performance information, change requests, project management plan updates, and project documents updates.
- Work performance information focuses on how well project communication is performed compared to what was planned.

PROJECT STAKEHOLDER MANAGEMENT

- You cannot control stakeholders, but you can monitor their level of engagement.
- Monitoring stakeholder engagement involves **monitoring overall project stakeholder relationships** and adjusting strategies and plans for engaging stakeholders as needed.
  - The project management plan, project documents, work performance data, enterprise environmental factors, and organizational process assets are important **inputs** to this process.
  - The project team uses data analysis, decision making, data representation, communication skills, interpersonal and team skills, and meetings as the **main tools and techniques.**
  - The **outputs** of monitoring stakeholder engagement are work performance information, change requests, project management plan updates, and project document updates.
- On some projects, stakeholder engagement is monitored by *making them key members of the project team*.
- It's important to understand the needs and desires of different types of customers as well.

PROJECT RISK MANAGEMENT

- Monitoring risks involves **tracking identified risk events and implementing the risk response plans,** while continuing to identify and analyze new risks and evaluate risk process effectiveness.
  - Project risk management does not stop with the initial risk analysis.
  - Identified risk events may not materialize, or their probabilities of occurrence or impact may diminish or increase.
  - Similarly, new risk events are normally identified as the project progresses.
- **Newly identified risk** events need to go through the same process of analysis and monitoring as those identified during the initial risk assessment.
- Project teams sometimes use **workarounds**— unplanned responses to risk events—when they do not have contingency plans in place.
- Data analysis (i.e., technical performance measurement and reserve analysis), audits, and meetings are **all tools and techniques** for performing risk monitoring.
- **Outputs** of this process include work performance information, change requests, project management plan updates, project documents updates (especially updating the risk register), and organizational process assets updates.

PROJECT PROCUREMENT MANAGEMENT

- Controlling procurements involves managing procurement relationships, monitoring contract performance, making changes and taking corrective actions as needed, and closing out contracts.
- Several tools and techniques can help in contract administration, including expert judgment, claims administration, data analysis, inspection, and audits.
- In addition to work performance information, change requests, project management plan updates, project documents updates, and organizational process asset updates, key outputs of controlling procurements include closed procurements and procurement documentation updates.
- It is very important to **document all changes to the contract** and communicate those changes to all affected stakeholders.
- Project team members must be aware of the **potential legal problems** of their not understanding a contract. Changes must be handled properly for items under contract.
- Professionals in procurement management should be consulted as needed.
- It is critical that project managers and team members watch for constructive change orders. **Constructive change orders** are oral or written acts or omissions by someone with actual or apparent authority that can be construed to have the same effect as a written change order.

SAMPLE WRITTEN NOTICE OF A CLOSED CONTRACT

Global Construction, Inc. Contract Closure Notice
September 16

As described in our service agreement, this letter provides formal notice that the work you were contracted to perform for Global Construction has been completed. ABC Training developed a qualified sellers list containing thirty potential sellers and a report with one-page of key information on each seller. Payment is being processed based on the invoice provided.

Kristin Maur, the project manager, has provided the following performance assessment for the work provided:

"We were very pleased with the work of ABC Training. Members of the firm were professional, knowledgeable, and easy to work with. Global Construction depended on ABC Training to develop a qualified sellers list for this important project, and we were extremely happy with the results. On a scale of 1 to 10, you earned a 10!"

By: *Lawrence Scheller* Lawrence Scheller, Contract Specialist Date: September 16

Figure 8-18. Sample contract closure notice

APPLYING PROJECT MANAGEMENT PRINCIPLES TO MONITORING AND CONTROLLING PROJECTS

- Although several of the project management principles apply to monitoring and controlling projects, the one principle that stands out is embracing **adaptability** and **resiliency**.
- **Adaptability** is the ability to respond to changing conditions.
- **Resiliency** is the ability to absorb impacts and to recover quickly from a setback or failure.
- As described throughout this chapter, it's important for project teams to expect changes, plan for them, and be adaptable and resilient in responding to them. Below are a few suggestions to help teams embrace adaptability and resiliency:

  1. Try to build **diverse teams with broad skill** sets and experiences
  2. Emphasize the importance of **continuous learning and improvement**
  3. Encourage **open conversations** and new ideas
  4. Provide and receive **feedback** often
  5. Focus on meeting **desired outcomes** instead of just providing planned deliverables

MONITORING AND CONTROLLING AGILE/HYBRID PROJECTS

- **Hybrid projects can use any** of the predictive processes, tools, and techniques to monitoring and controlling mentioned so far in this chapter as well as any agile approaches they decide will help the project be successful.
- The Scrum Guide (2020) does not mention project control directly, but several events assist in monitoring and controlling as follows:
- **Daily Scrum**: "The purpose of the Daily Scrum is to inspect progress toward the Sprint Goal and adapt the Sprint Backlog as necessary, adjusting the upcoming planned work." Developers can also meet throughout the day to discuss adapting or **replanning** the rest of the sprint's work.
- **Sprint Review**: "The purpose of the Sprint Review is to inspect the outcome of the Sprint and **determine future adaptations**. The Scrum Team presents the results of their work to key stakeholders and progress toward the Product Goal is discussed. During the event, the Scrum Team and stakeholders review what was accomplished in the Sprint and what has changed in their environment. Based on this information, attendees collaborate on what to do next. The Product Backlog may also be adjusted to meet new opportunities."

At the sprint level, Scrum teams often use several artifacts to monitor progress include task boards or Kanban boards, burn charts, and velocity charts

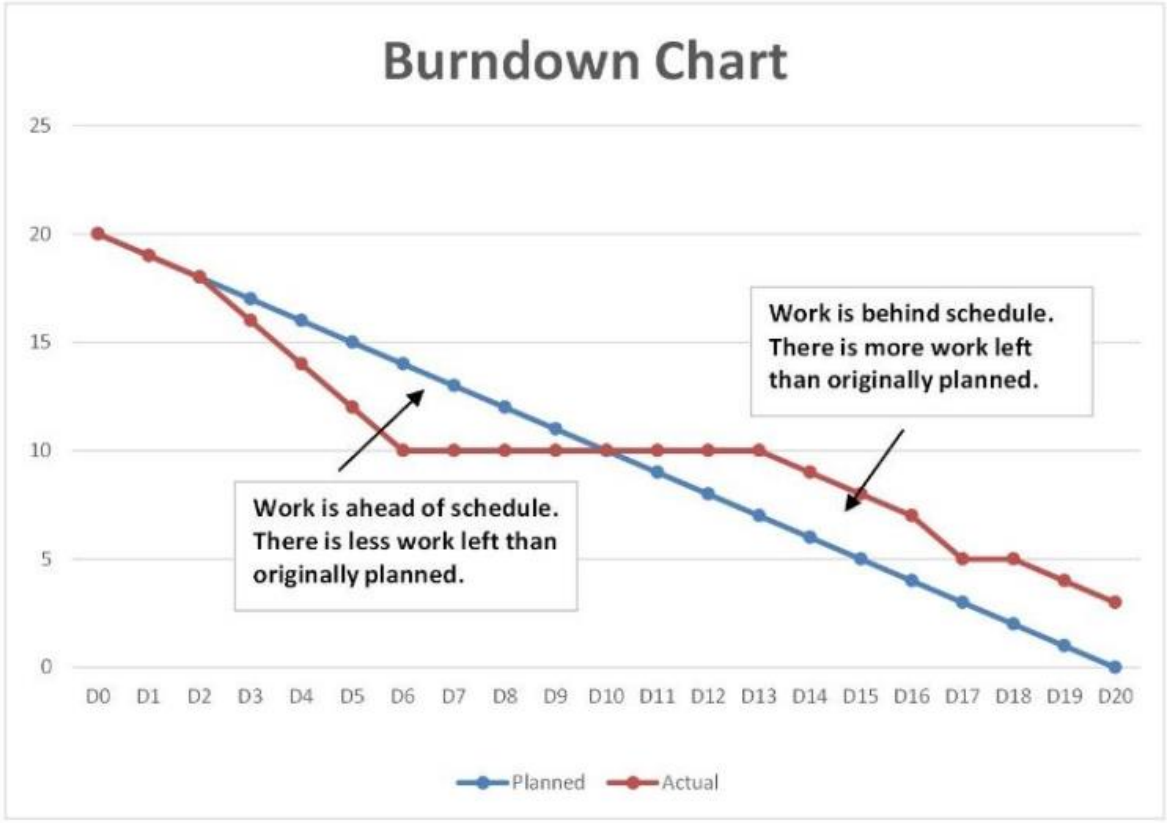
BURN CHARTS

- **Burn charts** show project team velocity.
  - "**Velocity** measures the productivity rate at which the deliverables are produced, validated, and accepted within a predefined interval."
- **Burndown charts** show the amount of work (number of tasks) ***remaining compared to the plan***. They are often used for each sprint and discussed during sprint retrospectives.
- **Burnup charts** show the amount of work (tasks) ***completed compared to the plan***. They can be used during each sprint, and they can also show progress for several sprints. Recall that the total amount of planned work often changes as the product backlog changes. Burnup charts show changes in planned work.
- **Combined burn charts** show ***how much work has been completed and how much remains.***

SAMPLE BURNDOWN CHARTS

- It is important to note that burn charts do not indicate which tasks have been completed.
- This means that a team can have a burndown chart that shows continued progress, but it does not show if the team has completed the most important tasks.
- Burn charts only provide an indication of trends rather than giving explicit indication of whether a team is delivering the right product backlog items if they do not complete all tasks

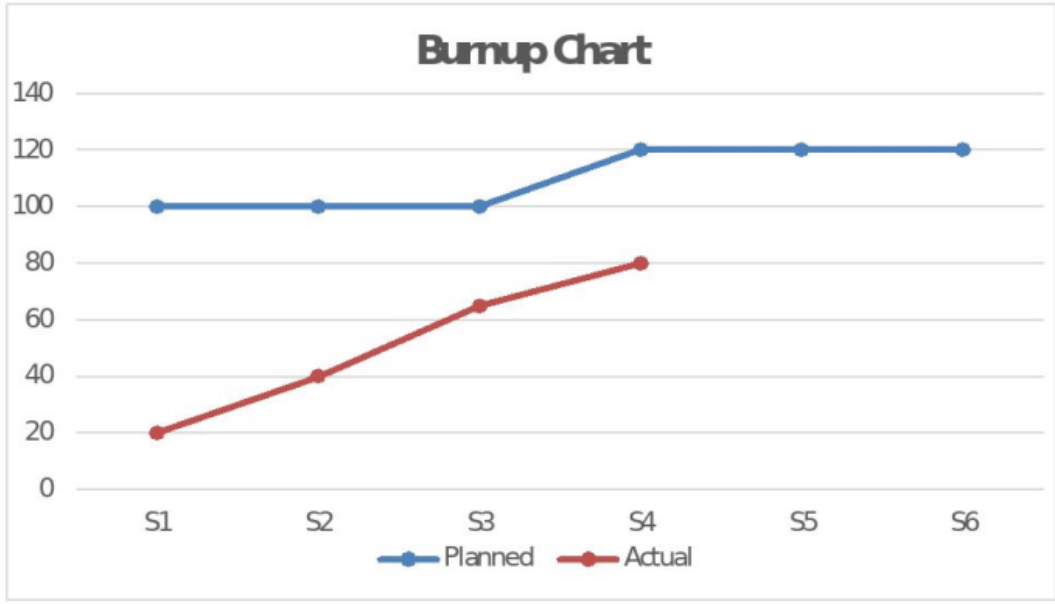
Days	Planned	Actual
D0	20	20
D1	19	19
D2	18	18
D3	17	16
D4	16	14
D5	15	12
D6	14	10
D7	13	10
D8	12	10
D9	11	10
D10	10	10
D11	9	10
D12	8	10
D13	7	10
D14	6	9
D15	5	8
D16	4	7
D17	3	5
D18	2	5
D19	1	4
D20	0	3



SAMPLE BURNUP CHARTS

Figure 8-19. Sprint burndown chart

Sprints	Planned	Actual
S1	100	20
S2	100	40
S3	100	65
S4	120	80
S5	120	
S6	120	



VELOCITY CHARTS

- Sprint teams use velocity to measure *how much work* they can complete in each iteration.
- It is widely used to help teams create accurate and efficient timelines.
- Sprint team velocity **is not constant**; rather, it varies.
  - Note that velocity charts are not intended to be a tool for monitoring the team.
  - They are most useful as a tool for release planning.
- The velocity chart is created after the first sprint and updated after each completed sprint.
- The velocity of the sprints helps managers to **calibrate the release plan**.
  - "There is no ideal velocity number; an upward trend is not necessarily a good thing and a downward trend is not always negative. But both will signal you to dig deeper into areas of the development process that can help you achieve efficiency while maintaining technical quality."

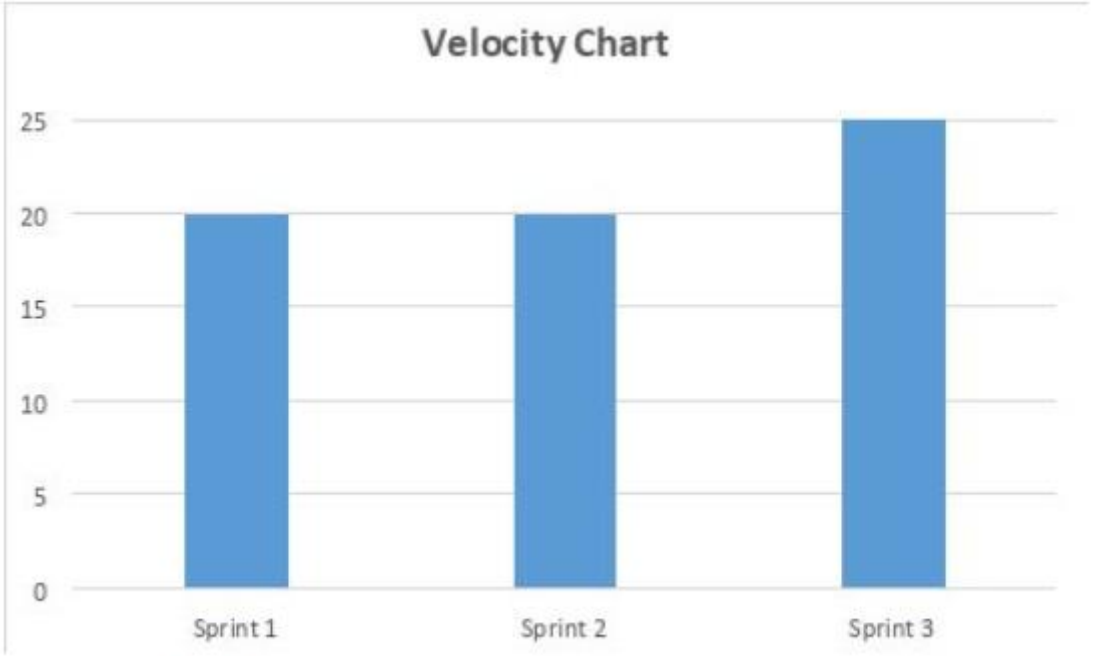


Figure 8-21 Sample velocity chart

CHAPTER SUMMARY

- Monitoring and controlling involves regularly measuring progress to ensure that the project is meeting its objectives and addressing current business needs.
- The project manager and other staff monitor progress against plans and take corrective action when necessary.
- Outputs common to several knowledge areas include change requests, work performance information, organizational process assets updates, project management plan updates, and project document updates.
- As part of **integration** management, project managers must monitor and control project work and perform integrated change control.
  - **Earned value management** is a project performance measurement technique that integrates scope, time, and cost data.
- Monitoring and controlling processes related to **scope** management include validating scope and controlling scope.
  - **Scope validation** means the customer or designated stakeholders have formally accepted the completed project deliverables.
- Controlling **costs** and schedules are common challenges for project managers.
  - **Earned value** management can aid in creating cost and schedule forecasts.
- Unique outputs of controlling quality include **quality** control measurements and verified deliverables.
  - Verified deliverables have been completed and checked for correctness as part of quality control.
  - There are many tools and techniques for controlling quality, including the seven basic tools of quality.

CHAPTER SUMMARY

- Controlling **resources** involves ensuring that the *physical* resources assigned to the project are available as planned and monitoring the planned versus actual resource utilization, taking corrective action as needed.
- Monitoring **communications** involves monitoring communications throughout the project life cycle to ensure that stakeholder information needs are met.
- Monitoring **stakeholder** engagement involves monitoring overall project stakeholder relationships and adjusting strategies and plans for engaging stakeholders as needed.
- It is important to reassess risks, and an important output of controlling **risks** is updating the risk register.
- Controlling **procurements** involves managing procurement relationships, monitoring contract performance, making changes and taking corrective actions as needed, and closing out contracts.
- Project managers should apply the project management principle of embracing **adaptability** and **resiliency** when monitoring and controlling projects.
- **Agile teams** also monitor and control their projects. Daily Scrums and sprint reviews are events that assist in monitoring and controlling on a regular basis. Burn charts, task boards, and velocity charts are artifacts that assist in showing progress to all interested stakeholders.